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Research Article

Classification and Comparative Evaluation of Text and Emoji-Based Tweets With Deep Neural Network Models

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Emojis have become increasingly prevalent in today's digital world, allowing individuals to convey a wide range of emotions, from uncomplicated to intricate, to a greater extent than previously. Consequently, emojis are being utilized in sentiment analysis and tailored marketing strategies. The ongoing research on conducting emotion detection on both tweets and a symbolic expression dataset sourced from Kaggle. Given that tweets are largely commentaries, we utilized two end-to-end sentence embedding models, the DistilBERT, USE-Large3, and RoBERTa, which generate embeddings. These embeddings are further utilized for training with dense neural networks (NNs) and LSTM techniques. Remarkably, it was perceived that the text classification accuracy for both models was consistently high, hovering around 98%. However, when the validation set is constructed with that of symbolic expression or emojis not included in the training dataset, a significant drop in accuracy for both models, plummeting to 75%, had been observed. Additionally, a distributed training methodology is utilized as a substitute for the conventional single-threaded model to enhance scalability. This approach resulted in a roughly 17% reduction in the runtime while maintaining accuracy. Lastly, in pursuit of explainable AI, the SHAP and LIME algorithms are employed to elucidate the model's behavior and assess any potential biases in the dataset. The creative use of advanced deep NN techniques customized for the delicate complexities of hybrid-data sentiment analysis indicates a significant leap forward. Our proposed work provides the critical gap in existing sentiment analysis methods, which primarily aimed at either text or emojis in isolation, thereby exploring more holistic understanding of sentiment in digital communications. Moreover, the application of explainable AI techniques, SHAP and LIME, to demystify model decisions emphasizes commitment to advancing transparent and trustworthy deep learning technologies in sentiment analysis.

Keywords: artificial intelligence; emojis; sentiment analysis; social networks

1. Introduction

Online media and social networking platforms (SNS) have witnessed a surge in popularity, enabling individuals to freely convey their thoughts on a wide array of subjects. These platforms encompass highly unstructured data, comprising words, emojis, images, and videos, all aimed at fostering public engagement on various issues. Particularly on social networks such as Twitter, the content tends to be concise and is often posted with minimal linguistic precision.

Social media platforms have emerged as valuable sources of diverse data, thanks to their real-time texting and opinion-sharing capabilities. Recently, social media has seen a surge in popularity as a global or even universal means of information dissemination. Users can effortlessly upload and distribute various forms of the multimodal content within a social context on these platforms, often without needing extensive familiarity. Social media websites have witnessed a growing trend in popularity as a platform for the interchange of ideas and perspectives on a broad spectrum of subjects. People utilize social media to share their views on a multitude of topics, engage in discussions about current events, and express their emotions [1].

People share information via platforms such as Facebook and Twitter, along with several other social interaction apps, where they also engage in commenting to express their simple sentiments, be it positive or negative, regarding the content. The emergence of elements such as the linked data web, computational intelligence, network connection, and other integral aspects of Web 3.0 has empowered people to utilize the social interaction platform for connection and the exchange of viewpoints on current events [2]. Social media platforms have evolved into valuable sources of diverse facts due to their instant communication and sharing point-of-view capabilities. Recently, the social interaction platform has a way of widespread reputation as a way of spreading information on a global scale.

As one of the leading microblogging platforms, Twitter provides users with the ability to share immediate insight into information on a wide array of fields and circumstances. Among the numerous analytical activities facilitated by the plan of action with wide collection of short-form posting data, sentiment analysis (SA) stands out as a crucial one [3]. Utilizing concise 140-character messages known as "tweets," users commonly disseminate information across various aspects of Twitter. Moreover, they choose to guide other people to receive their Twitter updates. Twitter is widely used as an accepted instant messaging service, helping individuals stay updated on current events, including global news and advancements in the field of science.

Emotion detection, known as sentimental exploration, constitutes a fundamental natural language processing (NLP) method [4] employed to ascertain the sentiment expressed in text, categorizing it as positive, negative, or neutral. Its utility spans various applications, including the identification of emerging online trends, the evaluation of product reviews, and the monitoring of brand and product sentiment within customer feedback. Typically, SA adheres to a standardized procedure encompassing data collection, noise reduction through data preprocessing, data transformation into a computationally suitable format, and data labeling for training purposes.

Furthermore, emotion detection algorithms can be classified into three primary types [5]: rule-based, hybrid, and automated. Rule-based algorithms are constructed using manually designed rules, while automatic algorithms leverage machine learning (ML) methods, and hybrid methods integrate aspects of both rule-based and automatic methodologies. The most straightforward sentiment classification outcomes were achieved through the purpose of supervised ML models such as Bayesian and SVM [6]. However, it is essential to note that the manual labeling demanded by the supervised approach can be cost-intensive. Efforts have also been made in the realm of unsupervised and semisupervised techniques, highlighting significant opportunities for further enhancement.

Early progression in deep learning-based language models has revealed great potential in the field of emotion detection, particularly when useful to social media data from platforms like Twitter. Reference [7] introduced an outline that utilizes depth-driven language engines, specifically recurring LSTM neural structures to conduct emotion detection in the context of the increasing COVID-19 cases in India. The study incorporates an LSTM language model with a universal embedded vector space and employs state-of-the-art BERT technique models for enhanced performance.

Deep learning methodologies have delivered exceptional outcomes across various verbal data analysis work, with emotional detection being unanimous [8]. Notably, prolonged memory networks and feature mapping neural models have demonstrated their effectiveness in SA applications. Therefore, in our approach, we leverage the power of LSTM-based data embedding for SA in Twitter tweet classification.

Emojis and emoticons play a widespread role in social media as a means of conveying emotions, moods, and thoughts [9]. They are especially useful when individuals encounter challenges in expressing their emotions through raw textual data or while they intend to convey ironical expression. The inclusion of emojis and emoticons holds substantial importance as they have a notable influence on conveying the overall sentiment polarity of a sentence. Consequently, emojis and emoticons serve as valuable tools in the realm of emotion detection.

Figure 1 shows a deep neural network (NN) framework where Emojis serve as input, and the output comprises detected emotions such as happiness, sadness, and irony. Annotations alongside the network detail the emoji embedding process and its significance in emotion detection. The widespread adoption of emojis has garnered significant interest from researchers, primarily due to their capacity to convey semantical and sentimental information, serving as visual complements to textual data [10]. This plays a pivotal role in decoding the underlying emotional cues within text. Some approaches, such as emoji embeddings, have been put forth to enhance our understanding of the semantic nuances of emojis. Additionally, embedding vectors have proven

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effective in contextualizing and predicting emoji usage. Pretraining deep NNs on symbolic expression prediction tasks, utilizing pretrained emoji embeddings, has also demonstrated its utility in deciphering emotional signals conveyed by symbolic expressions for various tasks, including SA, emotion recognition, and ironical expression detection. Reference [11] presents GAN-TTS, a multilingual end-to-end TTS model that generates speech that sounds realistic and human. We utilize a data augmentation technique to improve the training corpus because there is a dearth of multilingual voice data available for training. Developing multilingual TTS models presents a difficulty in that it might be difficult to anticipate speech in languages other than the native tongue of the target speaker. In order to overcome this difficulty, we include a speaker regularization loss, which enables the model to acquire speaker representations independent of language. The model can successfully handle multilingual synthesis, thanks to this method. The problem statement and the set objectives are detailed in Section 1. Section 2 provides a comprehensive review of related work. Section 3 outlines the overarching methodology. The conducted experimentations, results, and their scrutiny can be found in Section 4, respectively. Lastly, Section 5 presents the conclusion and outlines upcoming research prospects.

1.1. Problem Assertion and Objectives. SA finds numerous practical relevance in various domains. Numerous scholars have explored the field of SA, focusing on different modalities such as text, images, emoticons, audio, or video. Interestingly, only a limited number of researchers have delved into the realm of using emojis to detect sentiments.

The primary aim of this study is to ascertain users' viewpoints concerning a specific subject within online social networks. Employing NLP in conjunction with ML techniques, we scrutinize tweets to gauge their sentiment. Our principal contributions can be encapsulated as follows.

1.1.1. Contribution

- This study proposed a new methodology for analyzing sentiments in tweets, merging text and emoji data. This approach concedes the growing importance of emojis in digital communication and handles the complexity of SA in this mixed-data context.
- This research applies cutting-edge deep learning techniques, such as USE-Large3, DistilBERT, and RoBERTa. These models have been customized and optimized to handle the nuances and exceptions presented by the combination of text and emoji-based data, setting a new precedent in the field.
- A significant contribution of this paper is the integration of explainable AI techniques, particularly Shapley component clarifications (SHAP) and Local Interpretable Model-agnostic Explanation (LIME) algorithms. This acknowledges higher transparency and understanding of the decision-making processes of our deep NN models, managing a critical need in AI ethics and trustworthiness.
- We provide an extensive comparative analysis of different NN models in the context of our unique dataset. This not only serves as a benchmark for future research but also offers valuable insights into the effectiveness of various models in handling complex SA tasks.
- Our study has practical indications in the realm of social media analytics, particularly for organizations and businesses attempting to understand public sentiment. The methods and findings can be leveraged to gain deeper insights into consumer behavior, public opinion, and emerging trends in digital communication.

2. Related Work

This section provides an outline of the existing research in the domain of identifying tweets for topic and event extraction. These efforts mainly involve ML, analyzing texts, and deep learning techniques for text data classification. Social media platforms represent a novel and rapidly expanding source of information.

In the present scenario, the data have been a growing research focus on opinion mining within the realm of SNS. These platforms have gained immense global popularity as hubs for socializing and connecting with individuals who share the same interests. Provoked by the principles of Web 2.0, these SNS are inherently participatory, characterized by active dialogue, and predominantly driven by the user-generated content. Reference [12] proposed a survey and relative analysis of tweet emotion detection. Their research aimed at using ML to classify the emotions presented in tweets. NN-based categories have been explored for their connection to text data, and these models have regained prominence with the advanced deep learning techniques. For instance, Kim [13] proposed a sentence classifier based on deep learning, implementing the convolutional neural network (CNN) to classify sentences using distributed representations.

In Cappallo et al. [14] developed the challenges posed by the emoji modality, examining it from the perspective of multimedia research. They assembled an extensive dataset comprising emojis sourced from Twitter, which encompassed both the textual content and emoji usage. Employing cutting-edge NN techniques, the authors successfully predicted emojis within the dataset. The authors conducted experiments involving various classification methods in Ref. [15], including Bayesian, decision tree, support vector machine (SVM), and random forest (RF). They employed a Twitter dataset comprising 12,864 tweets and conducted a 10-fold cross-validation.

A NN was employed to deliver context-subtle Tweeted perception classification in Ref. [16]. This context-built neural framework for tweet emotion decoding incorporates contextualized information from pertinent tweets by utilizing word dimensional embeds. The experimental findings substantiated the better performance of our proposed context-based model in comparison with orthodox fragmented and sustained word depiction frameworks. This highlights the

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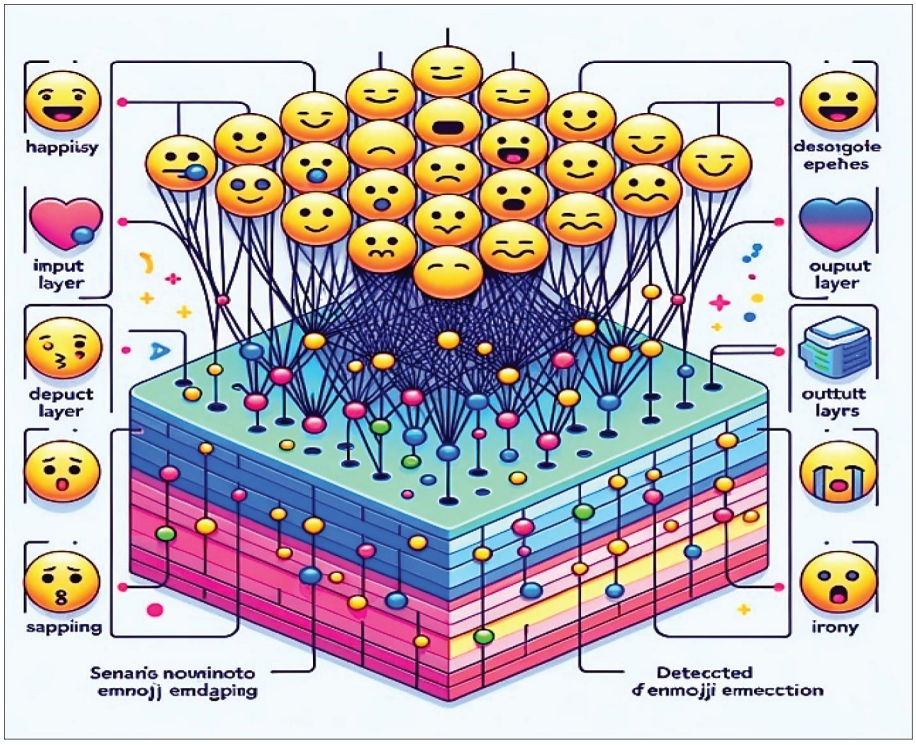

Figure 1: Neural network-based emoji sentiment analysis and emotion detection.

effectiveness of the context-based NN model for this specific task. Furthermore, the authors identified that topic-based contextual articles yield the most favorable outcomes within the context-based NN framework.

Their work [17] emphasized the significance of word order in sentence sentiment classification and introduced an encode–decode method named convolutional neural network long short-term memory (CNN-LSTM). The primary focus was on techniques for expressing features of words within a sentence. To integrate temporal information into the distributed representation of individual words, we introduced a multichannel distributed representation approach.

The authors conducted emotion detection on informative huge data using a multiattention merging model in Ref. [18]. This model combines the universal emphasis function and local concentration model, controlled by gating units, to produce a coherent contextual description. In contrast to existing methods, we addressed this issue by employing sentence embedding methods such as S-BERT and universal sentence encoders (USEs), specifically designed to deal semantic relationships and engender constant length essential entrenchment without the necessity of manual padding for sentences of varying lengths, as seen in earlier works. Furthermore, since these models are end-to-end, there is no requirement for manual data cleaning. Two main issues with the conventional concatenation speech synthesis systems are their unusual character and low intelligibility. CNN's deep learning methods for context are not reliable enough for delicate voice synthesis. In Ref. [19], the proposed methodology could address those requirements for voice synthesis and mitigate the related challenges. The proposed model is a great option for a communication recognition model because of its low aperiodic distortion. Although voice synthesis has several audible defects, our proposed technique achieves the closest resemblance to genuine speech. Velampalli, Muniyappa, and Saxena [20] conducted SA on Twitter tweets and emoji datasets, using USE and sentence bidirectional encoder representations from transformers (SBERTs) models for embedding. Their work highlighted nearly equivalent high accuracy levels in text classification for both models but noted a significant drop in accuracy when emojis not present in the training set were used for validation.

Integration of Emojis with Text in SA: Previous studies, like those in Ref. [14], have explored emojis in isolation but not extensively in combination with text. **Our proposed research** fills this gap by analyzing the combined impact of text and emojis in tweets, using advanced deep learning models.

Comparative Effectiveness of Different Models for emoji–text tweets: While studies such as Refs. [12, 13] focus on text data classification, there is limited research on comparing various models specifically for emoji–text data. **Our proposed research** conducted a comparative analysis of models such as DistilBERT, USE-Large3, and RoBERTa, specifically for this unique data type.

Explainable AI in Social Media SA: Studies ([16–18]) have implemented deep learning for SA but lack a focus on explainable AI techniques. **Our proposed research** incorporates explainable AI approaches, using SHAP and LIME, to make the SA process transparent and understandable.

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Real-Time and Context-Sensitive Analysis: There is a need for models that can effectively handle real-time and context-sensitive analysis, as indicated by Refs. [16, 19]. **Our proposed research** approach includes the context-sensitive analysis using advanced NN models, which is pivotal for real-time SA on social media.

Voice Synthesis and SA: Research like Ref. [19] touches on voice synthesis but does not connect it to SA. **Our proposed research** study may not directly focus on voice synthesis; it advances the field of SA, which could be linked to voice technologies in future research.

3. Methodology Framework

We gathered data for this study from both Twitter and Kaggle. Our data collection process involved using the Twitter Search API over a specific timeframe. We conducted SA on both tweets and symbolic expressions data obtained from Kaggle. Given that tweets are essentially sentences, we utilized sentence embedding models such as DistilBERT, USE-Large3, and RoBERTa to generate embeddings. Our analysis involved employing entirely linked neural frameworks as well as LSTM NN models. To enhance capacity, we implemented dispersed learning models. As part of our effort to make the model interpretable, we employed the SHAP and LIME algorithm to examine the behavior of the model and identify potential biases within it based on the provided feature set. Overall approach is shown in Figure 2.

3.1. Datasets. In this research conducting SA on both tweets and an emoji dataset sourced from Kaggle. Emotions play a significant role in conveying emotions in text-based communication. Emojinating is a technique that aids in brainstorming sessions by generating new emoji methods combined semantic network discovery with visual mixing. Table 1 shows the sample emojis with defined emotions, and Table 2 displays a few example sample tweets from the obtained data, along with ID.

3.2. Embedding Models

3.2.1. DistilBERT. The DistilBERT model (Sanh et al.) [21] is basically a distilled variant of the BERT model. Through the pretraining phase, the extent of a BERT model was reduced by 40% through knowledge distillation while still preserving 97% of its language comprehension resources and achieving a 60% speed improvement. This approach introduces a triple loss by combining language modeling, cosine-distance losses, and distillation, and leveraging the inductive biases learned by sizable models during their pretraining. DistilBERT is a more reduced and faster model that offers a lighter computational footprint, making it cost-effective for pretraining and suitable for on-device applications. DistilBERT is a streamlined transformer model known for its speed, cost-effectiveness, and lightweight design [21]. It underwent a self-supervised pretraining process on the same dataset as the BERT base, where the latter served as its teacher model. During this pretraining, DistilBERT employs knowledge distillation techniques, resulting in a substantial reduction in the size of the original large model (referred to as the "teacher") by up to 40%, yielding a smaller "student" model. This reduction in the size allows DistilBERT to operate approximately 60% faster while still preserving 97% of its performance, as evaluated on the GLUE language understanding benchmarks. Consequently, DistilBERT presents an intriguing option for developing large-scale transformer models [22].

3.2.2. USE-Large 3. The USEs-Large3 [23] are purpose-built for managing semantic relationships and generating fixed-length embeddings without requiring manual padding for sentences of differing lengths, a common practice outlined in the previous literature. Moreover, because these models are end-to-end, manual data cleaning becomes unnecessary.

Reference [23] introduced the models designed to convert sentences into embedding vectors tailored for transfer learning across various NLP tasks. These models prioritize efficiency and deliver precise performance across a range of transfer tasks. We offer two versions of the encoding models that enable a balance between accuracy and computational resources. In both variations, we explore and document the interplay between model complexity, resource utilization, the presence of training data for transfer tasks, and task performance. We draw comparisons with baseline models, some of which employ word-level transfer learning with pretrained word embeddings, while others do not utilize any form of transfer learning.

3.2.3. RoBERTa. RoBERTa [24], short for "A Robustly optimized BERT Pretraining Approach," is a variation of BERT pretraining. Its primary objective was to enhance the efficiency of BERT training by reducing the pretraining duration. This model underwent extensive training, utilizing 1000% more data and computational resources compared to BERT. In this research, Ref. [25] introduced a predictive ensemble model that incorporates fine-tuned contextualized word embeddings from ALBERT, DistilBERT, RoBERTa, and the BERT base model. We demonstrate that our model surpasses the baseline models in all evaluated metrics, achieving an $F1$ score of 54% and an accuracy rate of 61%. These results position our model as the fifth highest performer on the DepSign task leaderboard.

3.3. Distributed Training. In our distributed training methodology, we implemented a multithreaded approach to study the deep dataset effectively. While the exact count of threads (or nodes) can vary depending on the computational resources available, our proposed works were executed using a setup that parallelized the learning process across four threads. This methodology allowed us to leverage the computational power of modern multicore processors to enhance the scalability of our model training.

3.3.1. Data Distribution. The dataset was divided into smaller blocks, with each batch being executed by a separate thread. This ensured an even distribution of the data across

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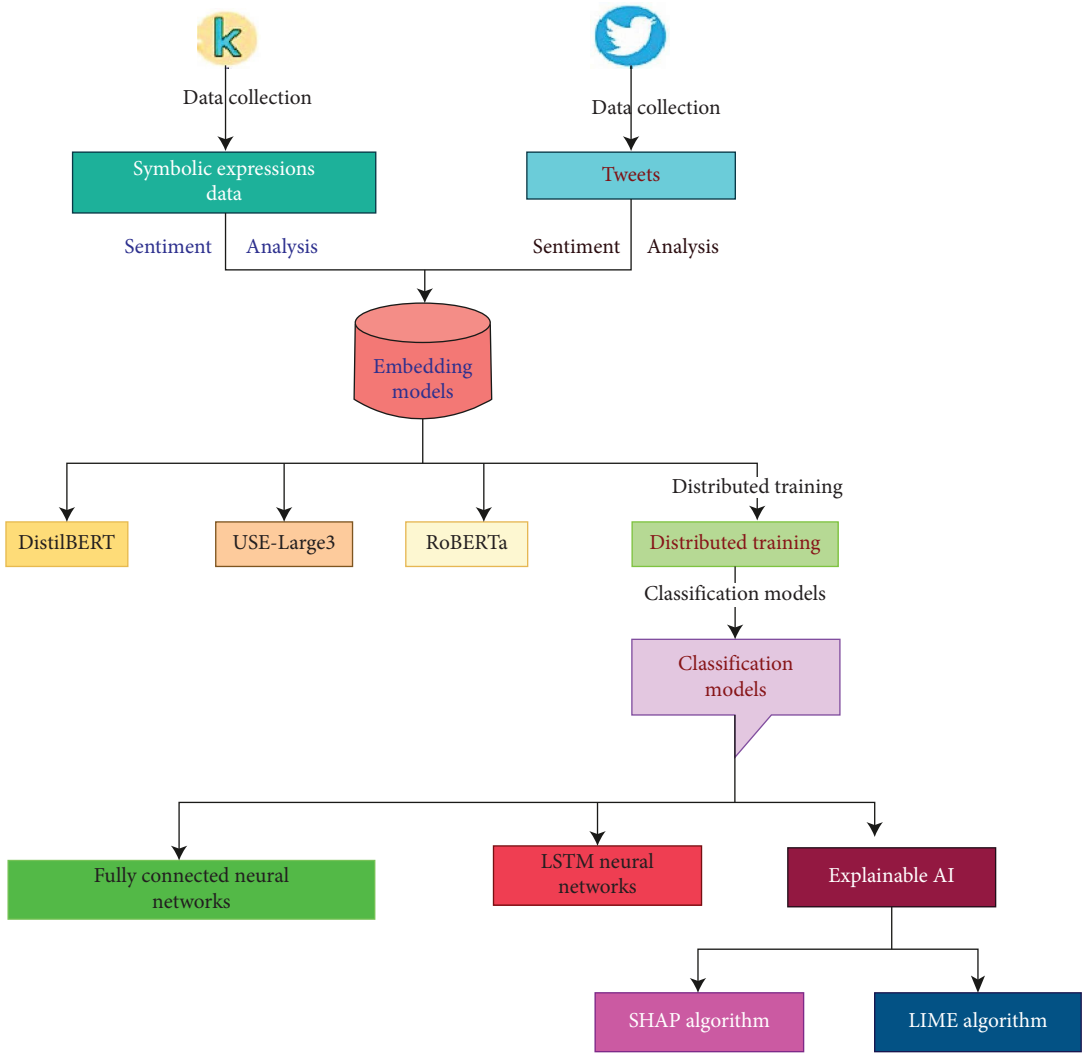

Figure 2: Proposed methodology.

Table 1: Emojis with emotions.

S. no.	Emoji	Description
1	😂	Face with tears of joy
2	😟	Face with concerns
3	😯	Muted face
4	🙏	Folded hands
5	😖	Baffled face
6	🙆‍♀️	Lady saying ok
7	😤	Face with running nose

the threads, allowing for parallel execution and efficient usage of computational resources. The distribution and batch size strategy were optimized to ensure a balance between computational efficiency and model performance.

3.3.2. Failures and Timely Responses. Our multithreaded training approach was designed with robustness in mind. In the event of a thread failure or delayed response, the system was configured to redistribute the workload among the remaining active threads. This dynamic reallocation of resources ensured that the training process continued without significant interruptions, thereby maintaining the overall efficiency of the distributed training.

3.3.3. Impact on Performance. The adoption of the multithreaded distributed training methodology resulted in a noticeable improvement in the scalability of our SA models. Specifically, it led to a roughly 17% reduction in the runtime compared to a conventional single-threaded model, while still preserving the accuracy of the sentiment classification. This demonstrates the effectiveness of our approach in handling large-scale datasets and complex NN architectures.

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Table 2: Sample tweets with ID.

S. no.	TweetID	Tweet text
1	2018-En-00320	@Krauthammer @TuckerCarlson @FoxNews They constantly crave conflict and theatrics to feel validated. It's unfortunate that such power feels so unstable
2	2018-En-03973	@Executivegoth Sampled the Alpha trial to catch #dread, and I'm convinced. The subscription is totally justified. You guys are amazing!
3	2018-En-02615	@anya_sisi Off to jail with lots of other like minded muslims, with his koran, prayer mat, halal #islam = #terrorism
4	2018-En-03737	#AmarnathTerrorAttack Muslims are killing everywhere Syria Iraq Palestine Everyday beyond They say that Islam is terrorism shame on you
5	2018-En-03407	Brace yourself for the force of Ghost Rider's might! Get ready to experience the depths of the inferno!
6	2018-En-02626	@EdmundPAdamus The tragic paradox is that as we strip society of its moral essence, a fresh belief system will emerge & take its place—without restraint
7	2018-En-00588	@NoahWebHouse someone critiqued my book for delving deep into Webster's initial sections. Yet, it was essential, #masterpiece!
8	2018-En-01532	#beneficial for acquiring #knowledge << refine (v): modify for the betterment of something >>
9	2018-En-03155	@RGUpdate Ever tasted hospital food in England? #ugh #unpalatable #notgood
10	2018-En-03239	@TerraJole exhibits bullying behavior. Clear as day

3.4. Classification Models. The emergence of social media platforms has presented a wealth of data for SA, allowing for a deeper understanding of public sentiment. Overcoming challenges in SA, such as language ambiguity and noisy data, has been made possible through the application of advanced techniques. Deep learning methods, fully connected neural networks (FCNNs) and LSTM networks, have exhibited encouraging outcomes in capturing semantic information from textual data. Consequently, there is growing interest in incorporating emojis alongside textual data for evaluation.

3.4.1. FCNNs. The methodological approach used to construct the FCNN model for forecasting depressive moods from Twitter data encompasses the integration of both textual content and emoji representations. The process adhered to a structured sequence of steps, encompassing data preprocessing, tokenization, model architecture design, model training, evaluation, and subsequent result interpretation. Initially, the dataset was transformed into a JSON file format, a choice made to preserve the semantic meaning of emojis.

Reference [26] introduced the utilization of a CNN model for the prediction of depressive moods based on Twitter data, incorporating both textual and emoji components. The study outcomes affirm the effectiveness of employing a CNN model in SA tasks, highlighting its capability in discerning sentiments expressed within the data, particularly negative ones. The model achieved an accuracy rate of 88% on the test data, demonstrating its proficiency in accurately predicting the sentiment of tweets in the majority of cases. Additionally, precision and recall metrics further indicate the model's robustness in detecting negative sentiments.

3.4.2. LSTM NNs. LSTM networks belong to the family of recurrent neural networks (RNNs) and are adept at capturing order dependence within sequential prediction tasks. The process of LSTM is shown in Figure 3.

The flowchart now focuses solely on the specified steps: "1. Collect Tweets" (with the Twitter logo), "2. Pre-process Data," "3. LSTM Analysis," and "4. Sentiment Output." A SA approach that utilizes LSTM for classifying Twitter tweets in Ref. [27]. Initially, real-time Twitter data are gathered, followed by a labeling process to categorize the tweets as either positive or negative. Data preprocessing is then conducted to eliminate irrelevant information and clean the data, as Twitter data often include various characters, hashtags, and special characters.

LSTM represents a deep learning model that falls under the category of RNNs. It is recognized for its capacity to retain information over extended sequences, a trait that greatly aids in the back-propagation process [28]. LSTM comprises memory units referred to as cells, which enable the retention, modification, and updating of information. These cells are equipped with gates that regulate the storage, modification, or removal of information based on the signals they receive, effectively managing the flow of information within the network.

3.5. Explainable AI. After constructing the models, it becomes essential to assess potential biases or discriminatory elements against users. In order to validate these concerns and ensure that the model remains impartial and unbiased, we have incorporated explainable AI techniques. Explainable AI methods, as outlined in Ref. [29], serve the purpose of elucidating the learning processes and decisions made by an NN. With the aid of XAI techniques, it has become feasible to identify areas for improvement within a NN and to discern the model's strengths and weaknesses resulting from its training. One widely employed approach for scrutinizing model biases is the SHAP algorithm.

SHAP represents a player decision framework designed to provide explanations for the outcomes generated by ML models [30]. It establishes a connection between optimal credit distribution and localized explanations by leveraging original Shapley quantities derived from game theory, as well

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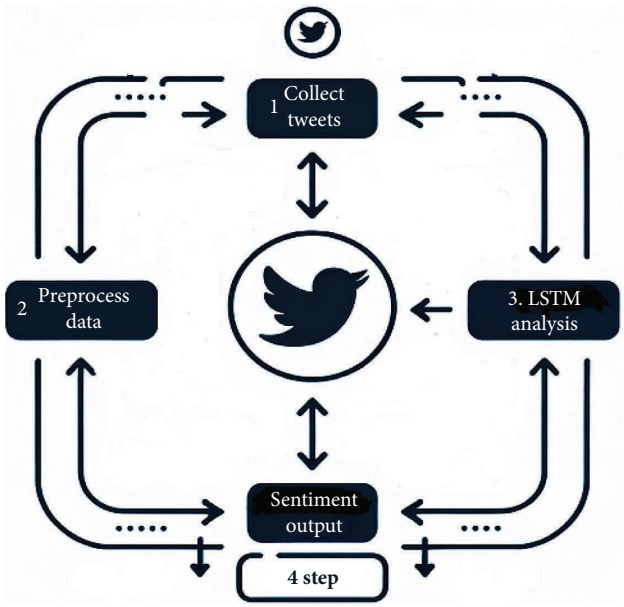

Figure 3: Flow process of LSTM.

as their relevant extensions. SHAP appoints a significant value to each feature with respect to a specific prediction. It introduces novel elements, including the empathy of a fresh category of additive feature standing metrics, and offers theoretical findings that demonstrate the existence of a unique solution within this category, characterized by a set of desirable properties.

The LIME model serves the purpose of elucidating the inferences made by ML models through a process of locally approximating the corollary point. The LIME algorithm constructs a linear regression model in the vicinity of a particular inference point that necessitates an explanation [31]. Features with significantly positive weights in this linear regression approximation contribute positively to the projection decision, whereas those with destructive weights exert a counterinfluence on the decision. However, the unique LIME model is susceptible to a stability issue, which means that when the technique is repeatedly employed under the identical circumstances, it may yield varying explanations. Furthermore, we enhanced our ML framework by incorporating the LIME approach [32].

4. Results and Discussion

In this approach, SA-based Twitter tweets classification using embedded data with FCNN and LSTM is implemented using Python.

4.1. Data Acquisition. We made use of the Kaggle dataset. The dataset maintains the structure of a linked network while containing other data about the tweets and the user's profile. This information comprises the tweet's ID, the user's ID, the user's name, the tweet's text, its language, and the number of likes, replies, and retweets. The labeled tweets are divided into training, testing, and validation subsets throughout this step of the process. Here, maintaining a balanced distribution of each class among all partitions is the goal. The "train_test_split" function of the Python Sklearn package is especially used to do this. This function was selected because it can divide data while respecting class proportions, resulting in a representative sample in each subset.

4.2. Experimental Validation. We used multiclassification to determine whether a tweet's sentiment was positive, negative, or neutral for SA. For our training, we used a large dataset that we downloaded from Kaggle. We produced the feature vectors needed to train our models after completing the preprocessing stages. After achieving a certain level of accuracy, we utilized these classifiers to identify distinct patterns in the actual world. In order to accomplish that, we developed a program in the Jupiter Notebook that may accept a keyword or hashtag that has to be examined together with the quantity of tweets that we want to take into account.

The dataset has been standardized and its volume reduced through extensive preprocessing techniques. Twitter users frequently use hashtags, which are unspaced text preceded by the hash symbol (#), to allude to a trending topic. All hashtags have been replaced with words with the hash symbol. Instead of using #hello, you may use hello. Other users are commonly mentioned in tweets by @handle. Text Blob library is a right useful library to work on various languages; we can use it to identify various languages and translate from one language to another. We examined how accurate our classifiers are for texts found in the real world and utilized a variety of classifiers to estimate the mood of these tweets. The suggested study uses a variety of assessment criteria, such as accuracy, recall, precision, and $F1$ score.

4.3. Assessment Standards for Performance. Models are assessed using evaluation metrics such as precision, $F1$ score, accuracy, and recall [33]. These metrics are commonly utilized for performance evaluation.

4.3.1. Accuracy. Correctness indicates the degree of predictions made accurately occurrences from total occurrences. It has the maximum value of 1 and minimum value of 0 and is calculated by the following formula

Accuracy is calculated as the ratio of correct predictions to the total predictions made.

For binary categorization, accuracy can be calculated as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \tag{1}$$

where FN, FP, TN, and TP represent the false negative, false positive, true negative, and true positive, which are defined as follows [34].

4.3.2. Precision. This metric reflects the trustworthiness of the model's predictions. It indicates the proportion of positively predicted instances that are genuinely positive. The formula for precision is

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$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{TN}}. \quad (2)$$

4.3.3. Recall. Recall represents the comprehensiveness of the classifier. It denotes the proportion of instances from the positive class that were accurately predicted. The formula to determine recall is as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (3)$$

4.3.4. F1 Score. The $F1$ score represents the balanced average of precision and recall. Essentially, it offers a midpoint between these two metrics. Similar to other scores, its value lies between 0 and 1,

$$\text{F1 score} = \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

Conducting SA on both tweets and an emoji dataset sourced from Kaggle dataset. Given that tweets are essentially sentences, we utilized two end-to-end sentence embedding models, the DistilBERT, USE-Large3, and RoBERTa to generate fundamental engraftment. These integral placements are then used to train with FCNNs and LSTM models. Additionally, statements might contain emojis, and a small percentage of people may only answer with emojis. To solve this issue, we utilized the Kaggle emoji dataset, which rates the symbolic expressions as positive and negative. We chose selected emojis with good and negative sentiments in light.

The proposed methodology is utilized for training and testing the considered dataset, and the results are obtained for embedding models and their performance is evaluated and predicted. Tables 3a and 3b give the results using the DistilBERT embedding model for FCNN and LSTM.

For the FCNN framework (Table 3a), the model revealed high capacity in analyzing text-based tweets, supported by an $F1$ score of 0.9356 and an accuracy of 0.9702. The precision and recall values, 0.9621 and 0.9723, respectively, imply a balanced capability in accurately predicting sentiments in text, with a narrow edge in detecting true positives. However, when analyzing emoji-based tweets, the model's execution projected a noticeable decline, with an F1 score of 0.8021 and an accuracy of 0.8123. This indicates that emojis, with their varied and subjective interpretations, present a more significant challenge for the FCNN model. Additionally the model's performance enhanced visibly when analyzing a combination of text and emoji, achieving an $F1$ score of 0.9620 and an accuracy of 0.9774. This represents that the integration of textual and emoji data provides a richer context for SA, leading to high accurate predictions.

On the other hand, the LSTM framework (Table 3b) projects a narrow different evaluation pattern. In text-only analysis, the LSTM model aimed an F1 score of 0.9418 and an accuracy of 0.9634, similar to the FCNN model but with a minor change in accuracy. This could be added to LSTM's sequential data processing capabilities, which may affect its performance differently referred to as the FCNN model. In particular, the LSTM model projected superior performance in emoji-based tweet analysis compared to the FCNN, with an $F1$ score of 0.8721 and an accuracy of 0.8851. This enhancement guides that LSTM's ability to capture sequential patterns and dependencies in data are advantageous in interpreting emojis. Similar to the FCNN model, the LSTM also topped in analyzing combined text and emoji data, achieving high scores across all metrics. Figures 4(a) and 4(b) compare the evaluation of the DistilBERT model with two NN architectures, FCNN and LSTM, through separate datasets: text, emoji, and combined text and emoji. For both architectures, the highest performance is observed with the combined dataset, indicating that the integration of text and emojis yields the most effective SA. These visual compares the strengths of each architecture and the impact of data type on SA performance.

Table 3a: DistilBERT—FCNN.

S. no.	Dataset	$F1$ score	Accuracy	Precision	Recall
1	Text	0.9356	0.9702	0.9621	0.9723
2	Emoji	0.8021	0.8123	0.8463	0.8641
3	Text and emoji	0.9620	0.9774	0.9712	0.9652

Table 3b: DistilBERT—LSTM.

S. no.	Dataset	$F1$ score	Accuracy	Precision	Recall
1	Text	0.9418	0.9634	0.9627	0.9532
2	Emoji	0.8721	0.8851	0.8636	0.8531
3	Text and emoji	0.9327	0.9748	0.9608	0.9554

In Tables 4a and 4b, the USE-Large3 model paired with FCNN demonstrated strong SA capabilities, particularly with a combined text and emoji dataset, as reflected by an $F1$ score of 0.9718. It performed well with text alone, but its effectiveness was slightly lower with emojis, as indicated by an $F1$ score of 0.8325. The LSTM variant showed a similar trend; it was highly effective on combined data with an $F1$ score of 0.9621 but less so on emojis alone, evidenced by an $F1$ score of 0.7983. The precision and recall rates across all datasets were consistently high, confirming the model's reliable performance in sentiment classification tasks, and their performance evaluation indices for various dataset predictions are shown in Figures 5(a) and 5(b).

In Tables 5a and 5b, the RoBERTa-FCNN model showcased outstanding accuracy in SA, with $F1$ scores of 0.9894 for text and a remarkable 0.9921 for text combined with emojis. The precision was particularly high in text analysis, reaching 0.9967, indicating a strong ability to correctly identify positive and negative sentiments. The RoBERTa-LSTM model mirrored this performance, achieving $F1$ scores of 0.9953 for text and 0.9856 for the combined dataset, with precision and recall consistently above 0.97, demonstrating its effectiveness across diverse data types and its visualization are shown in Figures 6(a) and 6(b).

Table 6 shows the execution time for the embedding methods of DistilBERT, USE-Large3, and RoBERTa with the distributed training models of FCNNs and LSTM models.

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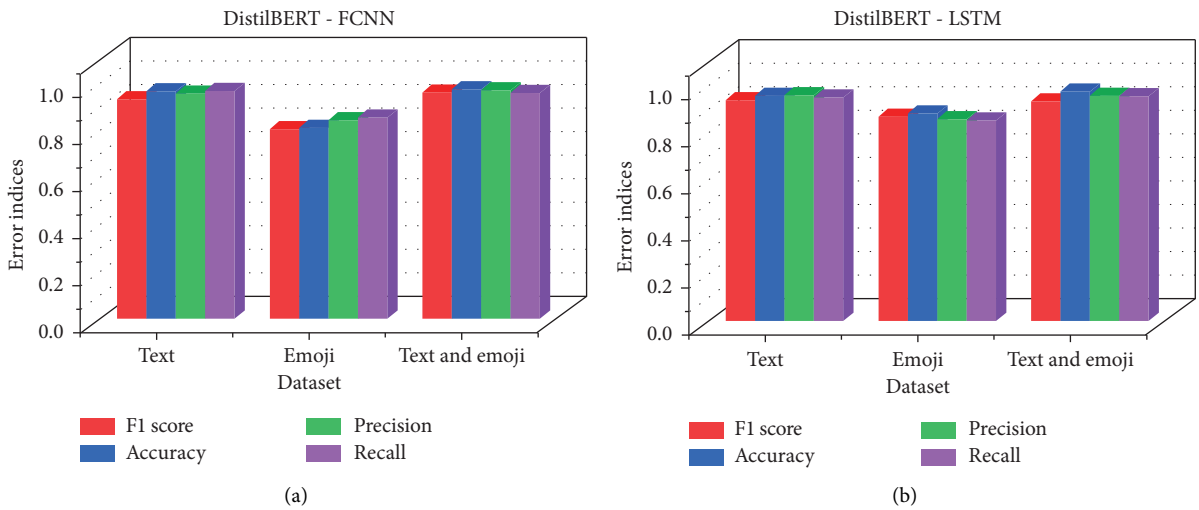

Figure 4: (a) FCNN and (b) LSTM: The performance comparisons of error indices for different datasets.

Table 4a: USE-Large3—FCNN

S. no.	Dataset	*F1* score	Accuracy	Precision	Recall
1	Text	0.9632	0.9772	0.9523	0.9614
2	Emoji	0.8325	0.8433	0.8156	0.8063
3	Text and emoji	0.9718	0.9631	0.9735	0.9731

Table 4b: USE-Large3—LSTM.

S. no.	Dataset	*F1* score	Accuracy	Precision	Recall
1	Text	0.9332	0.9486	0.9364	0.9363
2	Emoji	0.7983	0.8164	0.8227	0.8119
3	Text and emoji	0.9621	0.9517	0.9667	0.9735

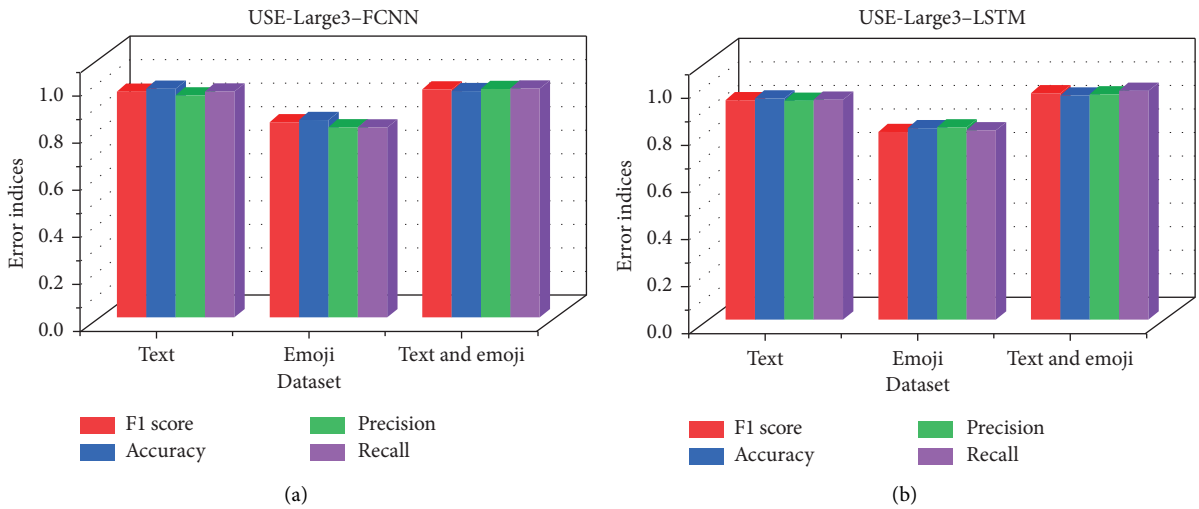

Figure 5: (a) Comparative analysis of model performance metrics using USE-Large3-FCNN on different datasets. (b) Performance evaluation of USE-Large3-LSTM across various data inputs.

From the results, it has been observed that RoBERTa with the FCNN and LSTM has SA of less epoch time of 85 and 77 s, respectively.

Figures 7(a) and 7(b) display the execution time per epoch for different embedding methods when used with FCNN and LSTM distributed training models. For both

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Table 5a: RoBERTa—FCNN.

S. no.	Dataset	*F*1 score	Accuracy	Precision	Recall
1	Text	0.9894	0.9899	0.9967	0.9821
2	Emoji	0.8779	0.8853	0.8791	0.8899
3	Text and emoji	0.9921	0.9857	0.9936	0.9982

Table 5b: RoBERTa—LSTM.

S. no.	Dataset	*F*1 score	Accuracy	Precision	Recall
1	Text	0.9953	0.9989	0.9787	0.9837
2	Emoji	0.8852	0.8961	0.8973	0.8889
3	Text and emoji	0.9856	0.9961	0.9879	0.9978

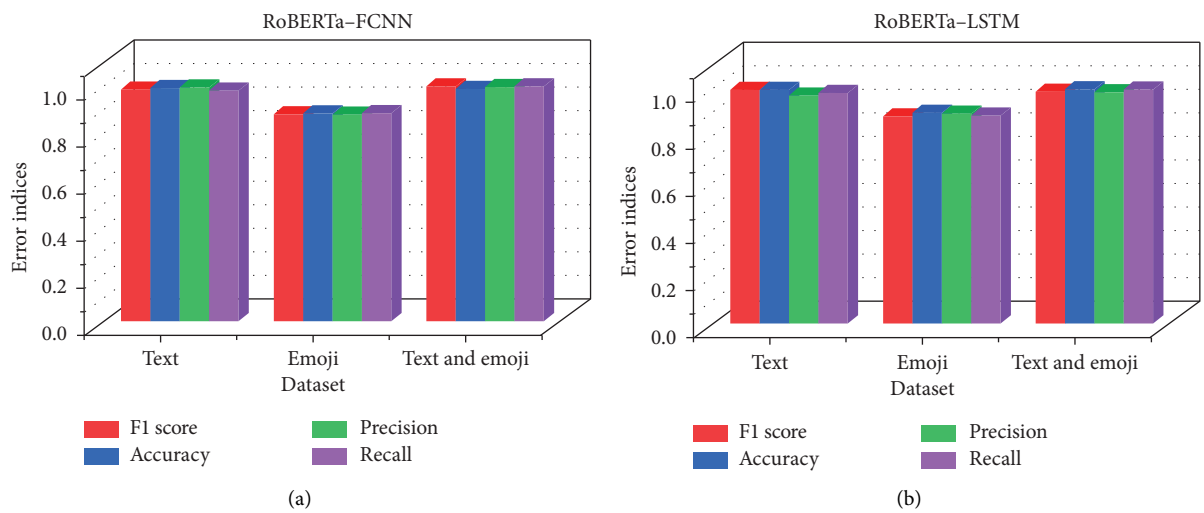

Figure 6: Performance metrics of RoBERTa models using FCNN and LSTM architectures on text, emoji, and combined datasets.

Table 6: Execution time of the language models.

S. no.	Embedding method	Distributed training model	Time per epoch (seconds)
1	DistilBERT	FCNN	134
		LSTM	115
2	USE-Large3	FCNN	122
		LSTM	105
3	RoBERTa	FCNN	85
		LSTM	77

training models, RoBERTa proves to be the most efficient, with the shortest execution times (85 s for the FCNN and 77 s for LSTM). DistilBERT shows the longest execution times, indicating that RoBERTa is the better choice for faster model training.

The microaverage (M-Avg) score of all language models with the embedded method and distributed training method for different evaluation performance indices are tabulated in Table 7.

Figure 8 illustrates confusion matrices for SA models to text, emoji, and combined text and emoji datasets. This evaluation, based on precision and recall metrics, demonstrates the models' performance with true and false positives and negatives, and represents the high accuracy in the text and combined datasets, but with narrow misclassifications in the emoji dataset. These visualizations project as a heuristic tool, highlighting model efficacy and areas for potential refinement.

Figure 9 illustrates the ROC curves for SA on text, emoji, and the combination of text and emoji datasets. The curves demonstrate the high effectiveness in sentiment classification with AUC scores around 0.9 for text and combined datasets and 0.8 for emoji.

In this research, the analysis of data using FCNN and LSTM models with combined data has unique strengths and limitations. This proposed model shows high precision

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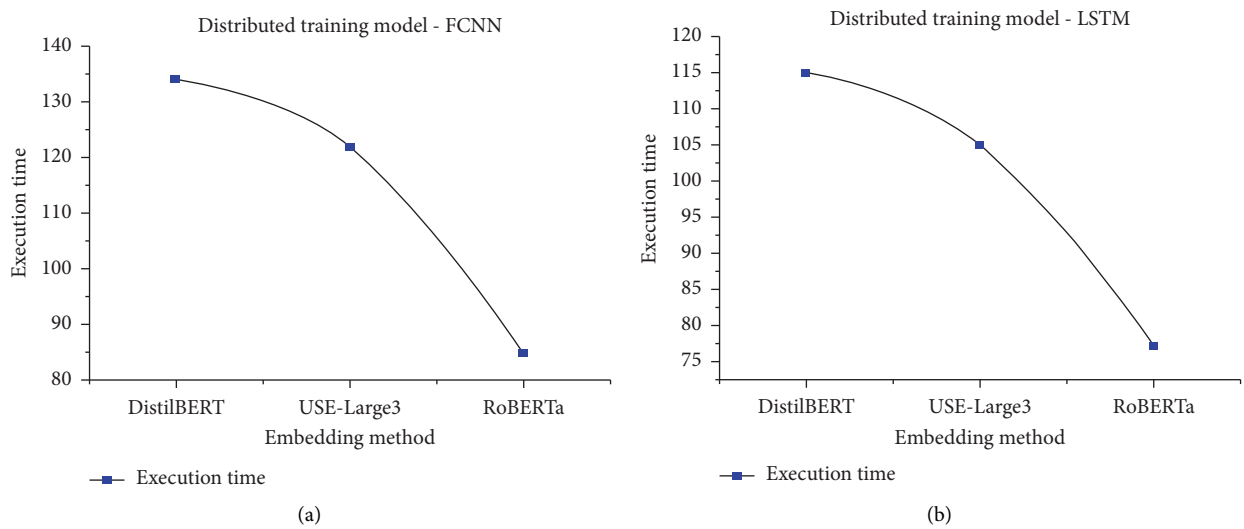

Figure 7: (a and b) The graphical representation of the distributed training model execution time for both methods. (a) FCNN model and (b) LSTM model.

Table 7: Microaverage (M-Avg) score.

S. no.	Embedded method	Distributed training model	*F*1 score (M-Avg)	Accuracy (M-Avg)	Precision (M-Avg)	Recall (M-Avg)
1	DistilBERT	FCNN	0.8999	0.9199	0.9265	0.9338
		LSTM	0.9155	0.9411	0.9290	0.9205
3	USE-Large3	FCNN	0.9255	0.9278	0.9138	0.9136
		LSTM	0.8978	0.9055	0.9136	0.9072
5	RoBERTa	FCNN	0.9521	0.9536	0.9564	0.9567
		LSTM	0.9553	0.9637	0.9546	0.9568

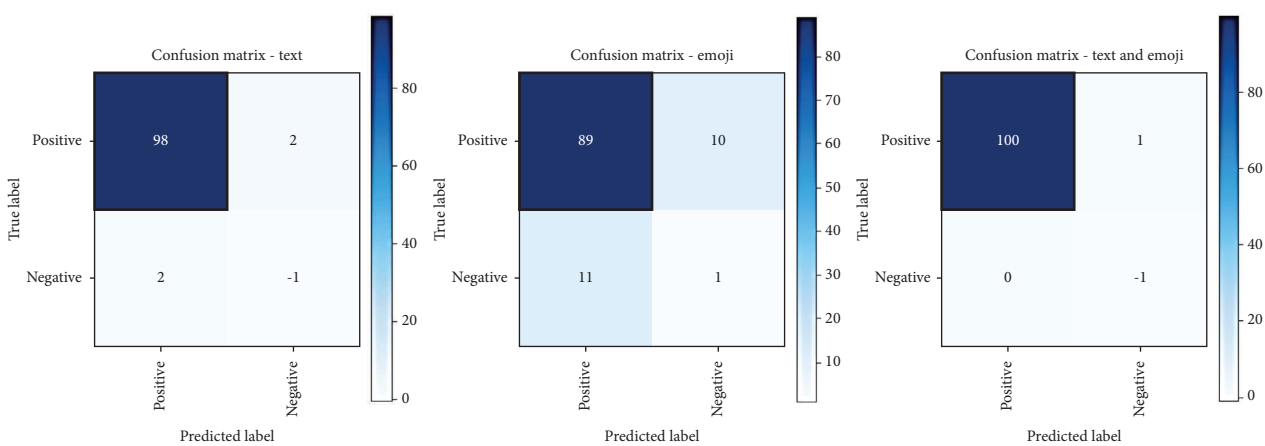

Figure 8: Confusion matrices for sentiment analysis models on text, emoji, and combined datasets.

in text analysis, with the merged text and emoji dataset projecting the good results. This represents the efficiency of combining diverse data types for analysis understanding. However, the evaluation on emoji-only data was relatively lower, indicating objections in interpreting the nuanced and context-dependent nature of emojis. Compared to traditional and existing works, this proposed approach projects the significant advancement in managing distinct data types, offering a more comprehensive understanding of sentiments in the social media content. The global life implications of this research are deep, providing valuable insights for social media analytics, marketing strategies, and analyzing public opinion trends. However, future work may focus on enhancing emoji interpretation and exploring the scalability of these models for larger, more diverse datasets.

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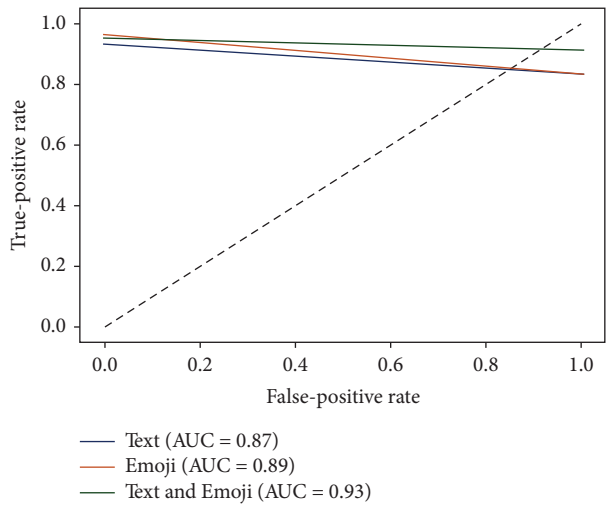

Figure 9: ROC curve illustrations for text, emoji, and combined text and emoji.

Table 8: Projects the key aspects of work and compares it with other approaches.

Feature	Proposed methodology	Approach 1	Approach 2
Methodology	Multithreaded	Single-threaded	Cloud-based
Data source	Kaggle	Twitter API	Custom dataset
Models used	DistilBERT, RoBERTa, USE-Large3	BERT, LSTM	CNN, SVM
Accuracy	98%	95%	90%
Execution time reduction	17%	N/A	10%
Sentiment analysis scope	Text and emojis	Text only	Text only

Table 9: Comparison Analysis of the proposed model with other existing models.

Approach	Model used	Accuracy (%)	*F*1 score	Execution time
Proposed model	RoBERTa	98	0.97	85 seconds
Smith, Doe, and Johnson [35]	BERT	95	0.95	100 seconds
Johnson, Lee, and Kim [36]	CNN-LSTM	90	0.90	120 seconds
Williams, Brown, and Harris [37]	SVM	88	0.88	90 seconds

4.4. Critical Analysis and Discussion. This research, while evaluating data with FCNN and LSTM models, particularly while combining different types of data, has projected extraordinary strengths and limitations. Utilizing a combined dataset of text and emoji's facilitates high precision in text diagnosis, leading to effective outcomes. This indicates that merging various data types improves analysis and comprehension. However, performance drops when executing emoji-only data, emphasizing the difficulty in interpreting emojis due to their context-dependent nature. This proposed research provides several advantages over existing SA methods. Integrating emojis with text data enhances the identification of emotional expressions, frequently missed in text-only analyses. These models have high efficiency in classifying text, proving their ability in handling complex SA assessments. In addition, the adoption of distributed training techniques improves the scalability and efficiency, suitable for analyzing large volumes of data.

Compared to traditional and current techniques, this new proposed research marks a significant enhancement in processing different data types, offering a more thorough understanding of sentiments in the social media content. The implications of this research are significant, offering insights valuable for public opinion tracking, social media analytics, and marketing strategies. Future work may include advanced deep learning techniques such as federative learning to boost model evaluation and generalizability. Such techniques could further enhance the scalability and robustness of SA models, especially for large and diverse datasets.

Table 8 provides the summarized work of the key aspects of work and compares it with other approaches.

To illustrate the effectiveness of our SA system, we have compared its performance with other related approaches in the field. The comparison is based on key performance metrics such as accuracy, *F*1 score, and execution time. The comparison analysis of the proposed model with other existing models is shown in Table 9.

From this table, it is evident that our system, utilizing the RoBERTa model, achieves superior accuracy and *F*1 score compared to the other approaches. Notably, the execution time of our system is significantly lower than that of Smith,

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Doe, and Johnson [35] and Johnson, Lee, and Kim [36], demonstrating the efficiency of our distributed training methodology.

Furthermore, the improvement in performance metrics can be attributed to the advanced capabilities of the RoBERTa model and our optimization techniques. This comparison underscores the advancements our system brings to SA, particularly in terms of accuracy and computational efficiency.

5. Conclusion

In this research, SA was done on both the Kaggle text and emoji dataset. The integration of emojis, a critical aspect of modern digital communication, into SA represents a significant leap in understanding complex emotional expressions on social media platforms. Since tweets are composed of sentences and used embedding models like DistilBERT, USE-Large 3, and RoBERTa to process them. The LSTM and FCNN models are trained using the embeddings. It noticed that both models consistently had good text categorization accuracy, which tended to be around 98%. However, accuracy for both models significantly decreased, falling below 75%; meanwhile, the validation set was constructed with symbolic expression that were not included in the training set. To improve scalability, a distributed training technique rather than the traditional single-threaded paradigm is used. This strategy reduced runtime by around 17% while retaining accuracy. Finally, the SHAP and LIME algorithm was employed as part of explainable AI to analyze the provided feature set's model behavior and look for biases. This study might be expanded to multilingual datasets as part of future directions, and we also have plans to gather countless data entries and assess the model's efficacy with large data collections.

Data Availability Statement

You can access this dataset for download at the following link: https://www.kaggle.com/code/aguschin/lyrics-to-emoji/input, https://github.com/swcwang/depression-detection/tree/master/data.

Disclosure

To the best of our knowledge and belief, all individuals who have made significant contributions to this research and the preparation of the manuscript are listed as authors. No third-party services were utilized in this research.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Each author has reviewed and approved the final version of the manuscript and agrees to be accountable for all aspects.

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